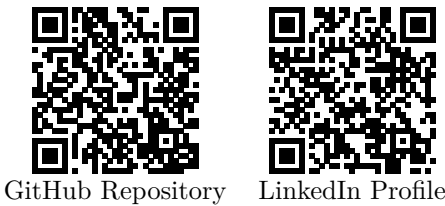


Environment Setup Manual

Concurrency & Distributed Systems — Lebanese University, Faculty of Engineering

Instructor: Dr. Mohamad Aoude | Language: Java 21+



Purpose

This guide prepares a student machine for the course labs. The repository is organized as independent Gradle projects:

Folder	Content
week1-lab/	Week 1 foundations, server waiting, benchmarking, Amdahl and Karp-Flatt exercises.
week2-lab/week2-day1-threads/	Thread creation, lifecycle, coordination, joins, futures, and race-condition basics.
week3-lab/week3-day1-race-monitors-lock-identity/	Main Week 3 lab on races, monitors, lock identity, states, and reentrancy.
week3-lab/week3-day1-locks-monitors/	Legacy Week 3 folder kept for compatibility with older references.
week4-lab/week4-day1-memory-visibility-immutability/	Visibility, volatile, atomicity, double-checked locking, immutability, and ThreadLocal preview.
week5-lab/week5-day1-juc-pools-backpressure/	java.util.concurrent, bounded queues, thread pools, atomics, read-write locks, and ThreadLocal cleanup.
Lectures/	Student-facing lecture PDFs. LaTeX source files are not published.
case-studies/	Course case studies.

Required Software

- Windows 10 or 11 with PowerShell
- Git
- Eclipse Temurin JDK 21 or newer
- Visual Studio Code
- VS Code Extension Pack for Java

Gradle does not need to be installed manually. Each lab includes a Gradle Wrapper script, so students should run `gradlew.bat` from inside the lab folder.

Step 1 - Install Tools

Open PowerShell as Administrator only for installation commands.

Install Git

```
winget install -e --id Git.Git --source winget
```

Install JDK 21

```
winget install -e --id EclipseAdoptium.Temurin.21.JDK --source winget
```

Install Visual Studio Code

```
winget install -e --id Microsoft.VisualStudioCode --source winget
```

Install Java Extensions

Close and reopen PowerShell, then run:

```
code --install-extension vscjava.vscode-java-pack
```

Step 2 - Verify the Environment

Open a new normal PowerShell window and verify:

```
git --version  
java -version  
javac -version  
code --version
```

The Java version must be 21 or newer. If PowerShell still sees an older Java installation, restart the terminal or the machine and check again.

Step 3 - Clone the Course Repository

Choose a workspace folder. The examples below use `C:\courses`; students may use another folder.

```
mkdir C:\courses  
cd C:\courses  
git clone https://github.com/maoude/concurrency-java-labs.git  
cd concurrency-java-labs  
code .
```

Step 4 - Run a Lab

Each lab is a separate Gradle project. Run commands from inside the specific lab folder. For a first environment check, run `classes`. Grading checks such as `studentCheck` are expected to fail while exercise TODO stubs are still empty.

Week 1

```
cd C:\courses\concurrency-java-labs\week1-lab  
.\gradlew.bat --version  
.\gradlew.bat classes
```

Week 2

```
cd C:\courses\concurrency-java-labs\week2-lab\week2-day1-threads  
.\gradlew.bat classes
```

Week 3

```
cd C:\courses\concurrency-java-labs\week3-lab\week3-day1-race-monitors-lock-identity  
.\gradlew.bat classes
```

Week 4

```
cd C:\courses\concurrency-java-labs\week4-lab\week4-day1-memory-visibility-immutability
.\gradlew.bat classes
```

Week 5

```
cd C:\courses\concurrency-java-labs\week5-lab\week5-day1-juc-pools-backpressure
.\gradlew.bat classes
```

After implementing an exercise, run:

```
.\gradlew.bat studentCheck
.\gradlew.bat clean test
```

Step 5 - Daily Student Workflow

1. Pull the latest course updates before starting work.
2. Open the relevant weekly lab folder in VS Code.
3. Read README.md, LAB_INSTRUCTIONS.md, EXERCISES.md, and CHECKLIST.md when present.
4. Implement only files under **exercises/** unless the instructor says otherwise.
5. Run `.\gradlew.bat studentCheck`.
6. Run `.\gradlew.bat clean test` before submitting.

```
git pull
```

Troubleshooting

Symptom	Fix
java is not recognized	Reopen PowerShell. If it still fails, reinstall JDK 21 and verify the JDK bin folder is on PATH.
Gradle asks for a different Java version	Install JDK 21 or newer. The labs use a Java 21 toolchain.
gradlew.bat is not found	Change directory into the weekly lab folder, not just the repository root.
VS Code cannot find Java	Install the Java Extension Pack, then reopen VS Code from the lab folder.
Student checks fail	Read the failing test name and message. Empty TODO stubs are expected to fail until implemented.

A correct setup lets you focus on concurrency behavior instead of tool problems.